CryptoTrend

ML-Powered Cryptocurrency Prediction Tool

MLOps

Djaferi Denis · Martinez Adrian · Nguyen Jason

The Problem

Why Crypto Trend Prediction?

- Crypto markets are highly volatile and trade 24/7
- Data shifts constantly => models degrade fast
- Need automated retraining & continuous monitoring
- **Perfect use case for MLOps practices**

$2.5T+

Global crypto market cap

24/7

Non-stop trading, data never sleeps

System Architecture

Data Ingestion

CoinGecko API
→ DVC versioning

→

Feature Eng.

RSI, MACD,
Moving Averages

→

Model Training

LogReg, Random
Forest and LSTM in
MLflow
Model registry

→

Testing & CI

PyTest + GitHub
Actions pipeline

→

Dashboard

Streamlit UI
Modelcomparison
Live predictions

End-to-end pipeline: from raw API data to live predictions in a web dashboard

Our Tool Stack

DVC

Version control for datasets as new crypto data arrives daily

MLflow

Track experiments, compare models, register the best performer

GitHub Actions

Automated CI/CD to run tests and validate on every push

Streamlit

Interactive dashboard for predictions and model monitoring

Ray Tune

Efficient hyperparameter search across model configurations

Alibi Detect

Detect data drift when market regimes shift

Team & Timeline

Data Pipeline

API ingestion, feature engineering, DVC setup

Model & Training

Baseline + LSTM, MLflow, Ray Tune

Deploy & CI/CD

Streamlit, GitHub Actions, drift detection

Timeline

W9	W10	W11-12	W13	W14
Setup project + data pipeline	Pitch (today)	Model + dashboard + CI/CD	Polish & rehearse	Final presentation

Progress & Next Steps

Completed

- Repo set up with DVC initialized
- Data ingestion from CoinGecko API
- Build Streamlit dashboard

Next Steps

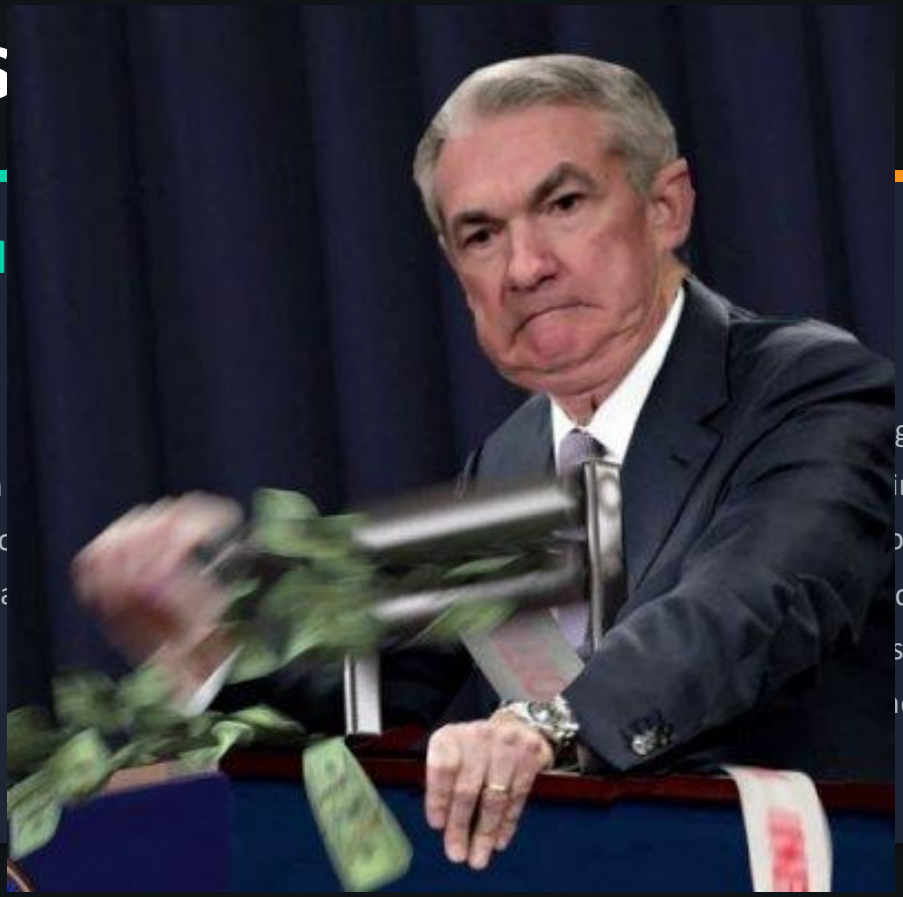
- Feature engineering pipeline
- Baseline model training & tracking
- Set up GitHub Actions CI/CD
- Integrate drift detection (Alibi)
- Possible Advancement: Demo-Trading
- Polish for final demo

Thank you · Questions?

Progress

Completed

- Repo set up with
- Data ingestion fr
- Build Streamlit da



g pipeline
ining & tracking
ons CI/CD
ction (Alibi)
s second approach
no

Thank you · Questions?